

20CS6T02 – Distributed Systems

UNIT 1: Introduction to Distributed Systems

Introduction

A distributed system is a collection of independent computers that appear as a single system to users. These systems work together to achieve a common goal.

This unit introduces the characteristics and advantages of distributed systems.

Detailed Explanation

Characteristics

Distributed systems provide resource sharing, scalability, and fault tolerance.

They enable multiple systems to work together efficiently.

Types of Distributed Systems

Examples include client-server systems and peer-to-peer systems.

Applications

Used in cloud computing, distributed databases, and large-scale applications.

Summary

This unit introduces distributed system concepts.

UNIT 2: Communication Models

Introduction

Communication is essential in distributed systems.

Detailed Explanation

Message Passing

Processes communicate by sending messages.

Remote Procedure Calls (RPC)

Allows calling functions on remote systems.

Applications

Used in distributed applications.

Summary

This unit explains communication models.

UNIT 3: Synchronization

Introduction

Synchronization ensures coordination among processes.

Detailed Explanation

Clock Synchronization

Ensures consistent time across systems.

Mutual Exclusion

Prevents conflicts.

Applications

Used in distributed computing.

Summary

This unit explains synchronization.

UNIT 4: Distributed File Systems

Introduction

Provides file access across systems.

Detailed Explanation

File Sharing

Allows access to files.

Consistency Models

Ensure data consistency.

Applications

Used in cloud storage.

Summary

This unit explains DFS.

UNIT 5: Fault Tolerance

Introduction

Ensures system reliability.

Detailed Explanation

Failure Types

Crash and network failures.

Recovery Techniques

Replication and backup.

Applications

Used in reliable systems.

Summary

This unit explains fault tolerance.